

Cloud Cost Estimator

Project Report

1. Introduction

Cloud computing allows organizations to use computing resources on demand. Managing cloud expenses is important for budgeting and resource optimization. This project demonstrates a simple cloud cost estimation model based on AWS pricing concepts.

2. Objectives

- Understand cloud service pricing.
- Estimate monthly infrastructure costs.
- Analyze the contribution of different services to total cost.
- Provide a simple reporting framework for cost planning.

3. Services Considered

Amazon EC2: Virtual servers used for application hosting.
Amazon S3: Object storage for files and backups.
Amazon RDS: Managed database service.
Data Transfer: Network traffic charges associated with cloud usage.

4. Cost Estimation Table

EC2 (100 Hours): \$5
S3 (200 GB): \$4.6
RDS (50 Hours): \$5
Data Transfer (100 GB): \$9

Total Estimated Monthly Cost: \$23.6

5. Benefits of Cost Estimation

Cost estimation helps organizations plan budgets, monitor spending, avoid unexpected charges, and identify opportunities for optimization.

6. Future Enhancements

- Multi-cloud support (AWS, Azure, Google Cloud).
- Dynamic pricing updates.
- Graphical dashboards.
- Automated cost optimization recommendations.
- Resource utilization analytics.

7. Conclusion

The Cloud Cost Estimator project provides a simple and practical method for understanding cloud infrastructure expenses. It demonstrates how cloud resources contribute to monthly costs and highlights the importance of cost management.